

Statistics II: Introduction to Inference

Problem set 5

1. Suppose $X_n = c + \frac{1}{n}Y$, where Y is a random variable with finite variance. Show that $X_n \xrightarrow{P} c$.
2. Let $X_n = \mathbf{1}_{A_n}$, where $P(A_n) = \frac{1}{n}$. Does $X_n \xrightarrow{P} 0$?
3. If $X_n \xrightarrow{P} X$, show that $X_n^2 \xrightarrow{P} X^2$.
4. Let $X_n = \frac{Z}{n}$, where $E[Z^2] < \infty$. Show that $X_n \xrightarrow{L^2} 0$.
5. If $X_n \xrightarrow{L^2} X$, does it imply $X_n \xrightarrow{P} X$? Prove or give reason.
6. Give an example where $X_n \xrightarrow{P} 0$, but $X_n \not\xrightarrow{L^1} 0$.
7. Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Show that the sample mean $\bar{X}_n$ is a consistent estimator of μ .
8. Let $T_n = \frac{n-1}{n}\bar{X}_n$. Is T_n a consistent estimator of μ ? Explain.
9. Let $X_n = (-1)^n$ with probability 1. Does X_n converge in probability? If yes, give the limit; if no, explain 'why'.
10. If $E|X_n|^2 \rightarrow 0$, then what is the limit of X_n in probability?
11. Let $X_1, \dots, X_n$ be i.i.d. with mean μ . Is the estimator X_1 consistent for μ ?
12. Let $X_n \sim \text{Poisson}(\frac{1}{n})$. Find the limit in probability of X_n .
13. Let $X_n \sim \text{Binomial}(n, \frac{1}{n})$. Find the limit in probability of X_n .
14. Let $X_n \sim \text{Exponential}(n)$ (rate = n). Find the limit in probability of X_n .
15. Let X_n have mean $\frac{1}{n}$ and variance $\frac{1}{n}$. Find the limit in probability of X_n .
16. Let X_n be such that

$$E[X_n] = 1, \quad \text{Var}(X_n) = \frac{1}{n}.$$

Find the limit in probability of X_n .

17. Let $X_n = \frac{1}{n} \sum_{i=1}^n Y_i$, where Y_i are i.i.d. with $E[Y_i] = 2$, $\text{Var}(Y_i) = 1$. Find the limit in probability of X_n .
18. Let $X_n = \frac{1}{n^2} \sum_{i=1}^n Y_i$, where Y_i are i.i.d. with $E[Y_i] = 1$, $\text{Var}(Y_i) = 1$. Find the limit in probability of X_n .
19. Let $X_n = \frac{1}{n} \sum_{i=1}^n Y_i^2$, where Y_i are i.i.d. with $E[Y_i] = 0$, $\text{Var}(Y_i) = 1$, and $E[Y_i^4] < \infty$. Find the limit in probability of X_n .
20. Let $X_n = \frac{1}{n} \sum_{i=1}^n (Y_i - \mu)^2$, where Y_i are i.i.d. with mean μ and variance σ^2 . Find the limit in probability of X_n .

Distributional Convergence, CLT, Continuous Mapping and Slutsky's Lemma

21. Let X_i s be independent discrete uniform random variables supported on $\{0, \dots, m\}$, $i = 1, \dots, n$. Find the asymptotic distribution of $\bar{X}_n$ as $n \rightarrow \infty$.
22. Consider X_i , $i = 1, \dots, n$, as in Q.21. Additionally, suppose Y_j , $j = 1, \dots, 20$, be i.i.d. from $\text{bin}(m, 1/2)$. Define $Z = (\sum_i X_i + \sum_j Y_j)/n$.
 - (a) Find the in-probability limit of Z , as $n \rightarrow \infty$.
 - (b) Find the asymptotic distribution of Z , as $n \rightarrow \infty$.
23. Consider X_i , $i = 1, \dots, n$, as in Q.21. Let Y_j , $j = 1, \dots, n$, be i.i.d. from $\text{bin}(m, 1/4)$. Find the in-probability limit of $\sum_i X_i / \sum_j Y_j$.
24. Let X_i , $i = 1, \dots, n$, be independent $\text{exponential}(\lambda)$ distribution. Find the in-probability and L^2 limit of $n^{-1} \sum_i \sqrt{X_i}$? What is the L^1 limit of $n^{-1} \sum_i \sqrt{X_i}$?
25. Let $X_1, \dots, X_n$ be i.i.d. $\text{uniform}(0, \theta)$. What is the in-probability limit of $\log(X_{(n)})$?
26. Let $X_n \sim \text{uniform}(0, 1 + n^{-1})$ independently across all $n = 1, 2, \dots$. Find the random variable X such that $X_n \xrightarrow{d} X$ as $n \rightarrow \infty$.
27. i. Let X be a continuous random variable, and $X_n \xrightarrow{d} X$. Prove that $X_n^2 \xrightarrow{d} X^2$ as $n \rightarrow \infty$.
 ii. Let T_n , $n = 1, 2, \dots$, and T be non-negative random variables, and $T_n \xrightarrow{p} T$.
 Prove that $T_n/(1 + T_n) \xrightarrow{p} T/(1 + T)$ as $n \rightarrow \infty$.
28. Let $X_1, \dots, X_n$ be i.i.d. $\text{normal}(0, \sigma^2)$. Find the asymptotic distribution of $|\bar{X}_n|$.
29. Let X_n and Y_n be non-negative random variables, and $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} 1$. Define

$$Z_n = \log(1 + X_n Y_n).$$

Find the distributional limit of Z_n .

30. Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c > 0$. Define

$$Z_n = \frac{\sqrt{X_n^2 + Y_n^2}}{Y_n}.$$

Find the distributional limit of Z_n .

31. Let $Z_1, Z_2, \dots$ be a sequence of random variables, and suppose that $n = 1, 2, \dots$, the distribution Z_n is as follows

$$P(Z_n = 0) = 1 - \frac{1}{n}, \quad \text{and} \quad P(Z_n = n^2) = \frac{1}{n}, \quad \text{for } n = 1, 2, \dots$$

Show that

$$\lim_{n \rightarrow \infty} E(Z_n) = \infty, \quad \text{and} \quad Z_n \xrightarrow{p} 0, \quad \text{i.e., for any } \epsilon > 0, \quad P(|Z_n| > \epsilon) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

[Note: This example shows that the sufficient condition for consistency of a sequence of estimators is not *necessary* (i.e., the converse is not true).]

32. Let $X_1, \dots, X_n$ be a random sample from some distribution with expectation μ , variance σ^2 and finite fourth order raw moment μ'_4 . Determine the sequence of real numbers $\{a_n\}$ and the random variable Z such that the sequence of random variables $Z_n = \sum_{i=1}^n X_i^2 / \sqrt{n} + \bar{X}_n^2$ satisfies $Z_n - a_n \xrightarrow{d} Z$. What is the distribution of Z ?